

Note that Artificial Intelligence and Machine Learning aren't two disparate fields of computing, in fact they are closely related.

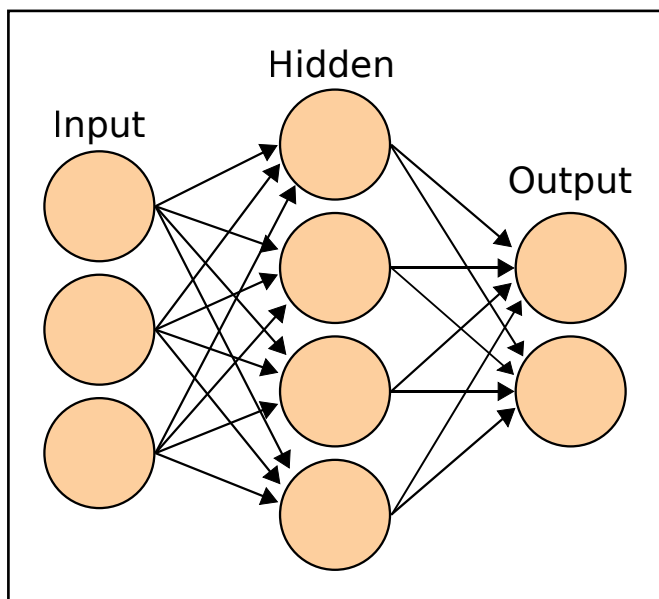
So when we speak of Machine Learning, we need to likewise account for the differences between humans and computers. Before we talk about what is learning for a computer, let's look at what learning means for humans. Learning is about gaining knowledge and skills through study or experience. Obviously this isn't exactly what you'd expect a computer to do, but Machine Learning does come quite close.

What is Machine Learning?

Machine Learning is about creating computer algorithms that can learn how to make optimal decisions based on models created by processing large amounts of data.

A popular example of this that you definitely would have heard of and encountered is the spam filter in email (and other places as well). There is no "formula" for calculating which email is spam and which isn't. So how does a computer know, how does it calculate the worth of an email?

In this particular case, there is no dearth of data available about what constitutes a spam email and what constitutes a genuine email. People provide this data themselves by marking an email as spam, and less often marking an email as not spam.



Neural networks sound complicated, but are easy to grasp once you know the basics of machine learning. In the above representation, the circles in the input column represent elements of the input, for example bank balance and salary, the lines represent their weights, there can be multiple hidden layers, that apply more weights, and the output is the result.

The spam filter uses this increasing database of spam and non-spam emails to develop a model of what a spam email is, and what it isn't. For instance, a tampered or missing email header might directly qualify an email as spam no matter what its contents and an email from a valid contact might automatically pass through the spam filter even if it is quite "spammy".

In fact the above example falls into a type of Machine Learning called Supervised learning, which we will cover in greater detail later.

Here are a few more examples that might give you an idea of what machine learning is:

- **Shopping recommendations on websites like Amazon:** These are called recommender systems and they look at data from millions of users to recommend products to users based on their browsing / purchase history.
- **Optical character recognition:** These systems can convert scanned documents into editable text by detecting characters in the scanned image. Usually these software are pre-trained to recognise scanned documents, but will improve with user guidance.
- **Voice recognition:** Similar to above, voice recognition technologies — such as the ones built into phones these days — include a model for voice recognition that the computer learns by learning relationships between voice and text. Often such software will continue improving by learning the users specific voice better.

In all these examples, data is a key feature, and the more you have the better. However there is more to it than just throwing more data at a problem. Let's look a bit at what training / learning is when it comes to machines.

Training with Data

To understand training / learning from data, let's take the example of a bank developing an algorithm to determine whether to approve a credit card request or not. There is no straightforward formula for this, but perhaps machine learning can help?

The data we can supply to the algorithm is all the data we have about the customer from the form they filled. We know their salary, their home address, their bank balance, whether they own the house they live in etc. We want the algorithm to look at this and tell us whether to approve or deny a credit card.

To train this system we will give it data we already have. We give it the above information about previous customers who we already issued a credit card, with an added piece of data that says whether the card should have been approved or denied. The bank can base this information on those people who regularly paid their credit card dues, and those who defaulted.

Looking at this data, the algorithm will then try to determine where to draw the line between a customer that should be given a card and the customer that shouldn't be given one. It will build up a formula composed of the different factors in consideration, such as the salary, and whether their house is owned or rented, and their bank balance. It will keep varying the weights applied to these parameters until it gets to a formula that gives the correct result for all the training data.

This isn't without problems though, and it comes down to the concepts of underfitting and overfitting.

Underfitting is what you end up with if you just don't have enough data to get the system to the point where it can make good enough decisions. For instance, if we have no information about the salary of the applicant, chances are we will not have enough data to make a decision. There is little you can do in this case then to collect more data.

It is also possible that the machine learns very well from the training data, and gets a perfect result, however, it utterly fails when applied to real world data. The formula the algorithm develops has learned to fit the training data so close that it